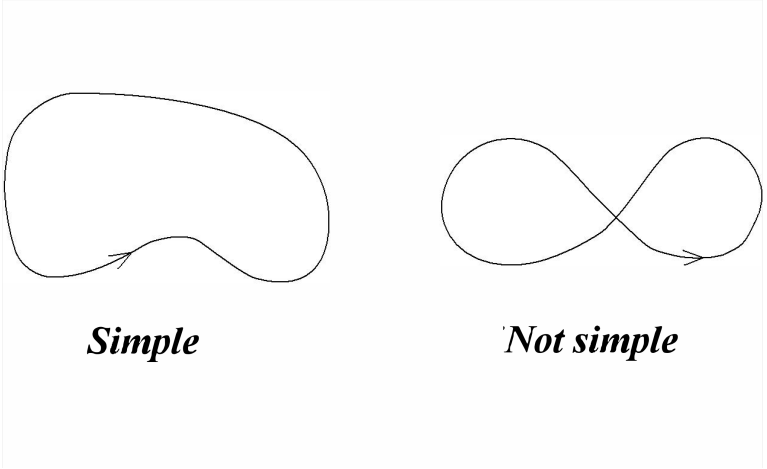
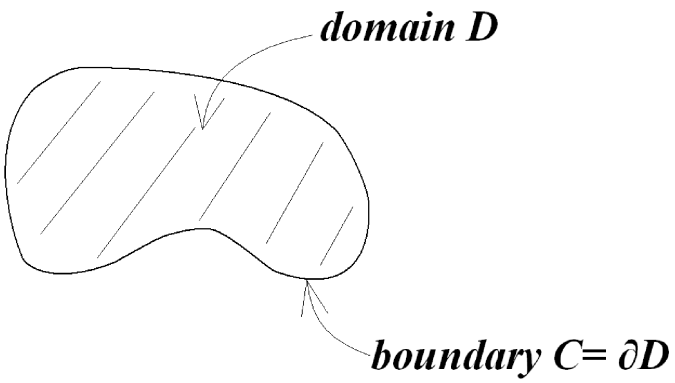
6. Integral Theorems

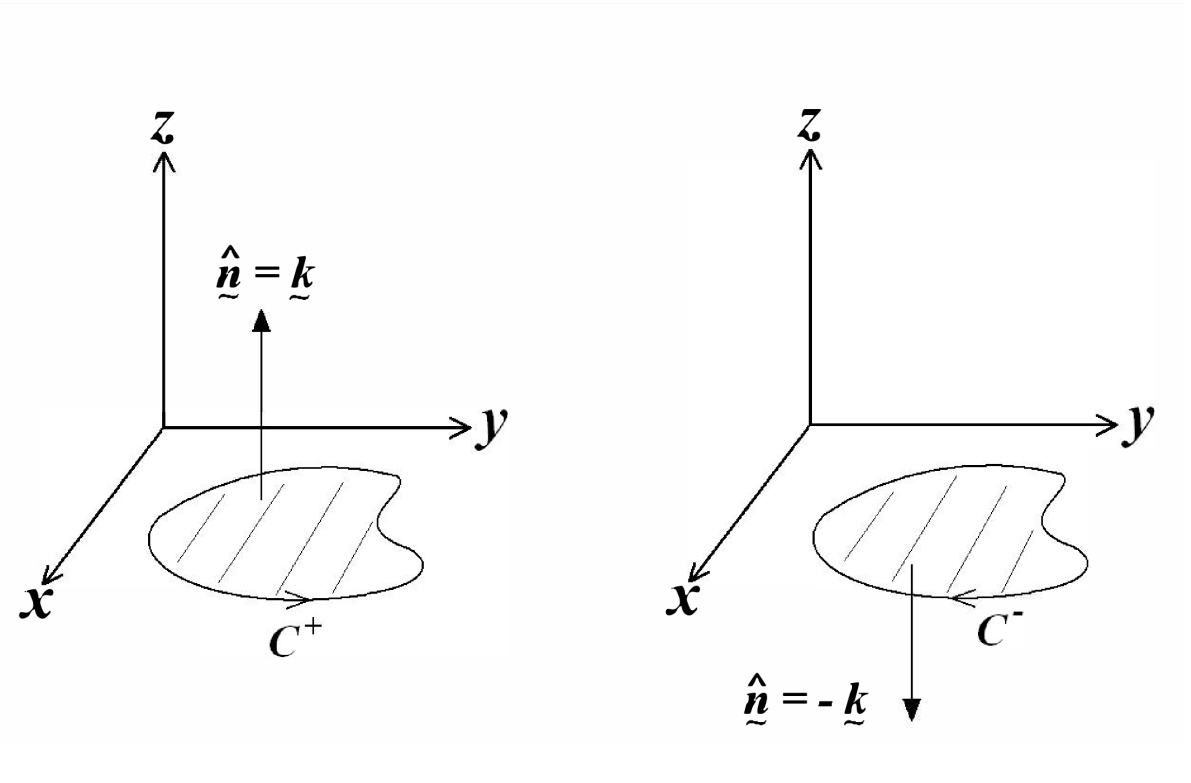
6.24. Oriented closed curves in the xy-plane

A curve C is simple if it does not intersect itself anywhere between its endpoints.



A simple closed curve C is positively oriented if the region D enclosed by C stays on your left as you move along the curve in the direction of the orientation.



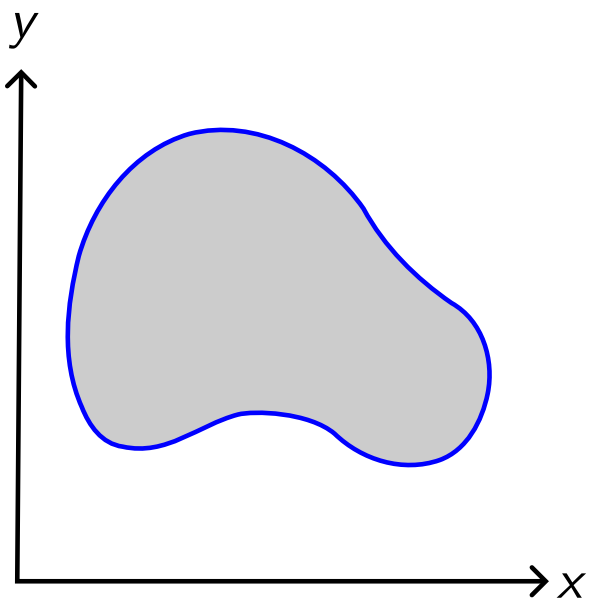


- The curve on the left is positively oriented (anticlockwise)
- The curve on the right is negatively oriented (clockwise)

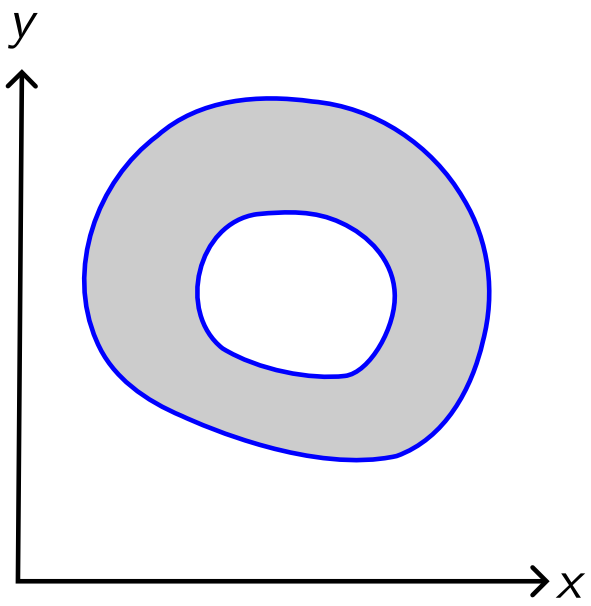
We choose $\hat{\mathbf{n}}$ so that it is the normal to the xy-plane related to the orientation by the right-hand rule.

A region D in the plane is connected if any two points in D can be joined by a piecewise smooth curve entirely in D.

A region D in the plane is simply connected if it is connected and every simple closed curve in D encloses only points that are in D.



simply connected



connected but not simply connected

Green's theorem in the plane

Theorem 24.1 (Green's theorem)

Let D be a simply connected region in the xy-plane bounded by a piecewise smooth simple closed curve C with positive orientation. If $P(x, y)$ and $Q(x, y)$ are C^1 functions on an open region containing D, then

$$\int_{C=\partial D} (P\,dx + Q\,dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy$$

Vector form of Green's theorem

Write $\mathbf{F} = P(x, y)\,\mathbf{i} + Q(x, y)\,\mathbf{j}$, a C^1 vector field on D. Then the expression in Green's theorem can be rewritten as

$$\int_{\partial D} \mathbf{F} \cdot \, d\mathbf{c} = \iint_D (\boldsymbol{\nabla} \times \mathbf{F}) \cdot \mathbf{k} \; dx\, dy = \iint_D (\operatorname{curl} \mathbf{F} \cdot \mathbf{k}\,)\, dx\, dy$$

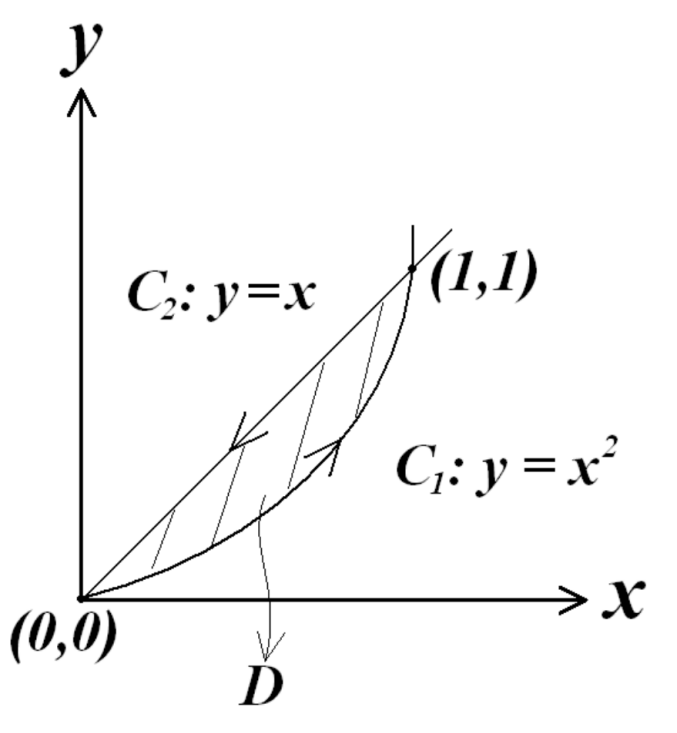
We check that the double integrals are equal.

Example 24.2

Let $\mathbf{F}(x, y) = (xy^2,\ y + x)$ and consider the region D bounded by

$$y = x^2,\ y = x,\ (x, y \geq 0).$$

Compute $\int_{\partial D} \mathbf{F} \cdot d\mathbf{s}$ using Green's theorem, and also directly as a contour integral.

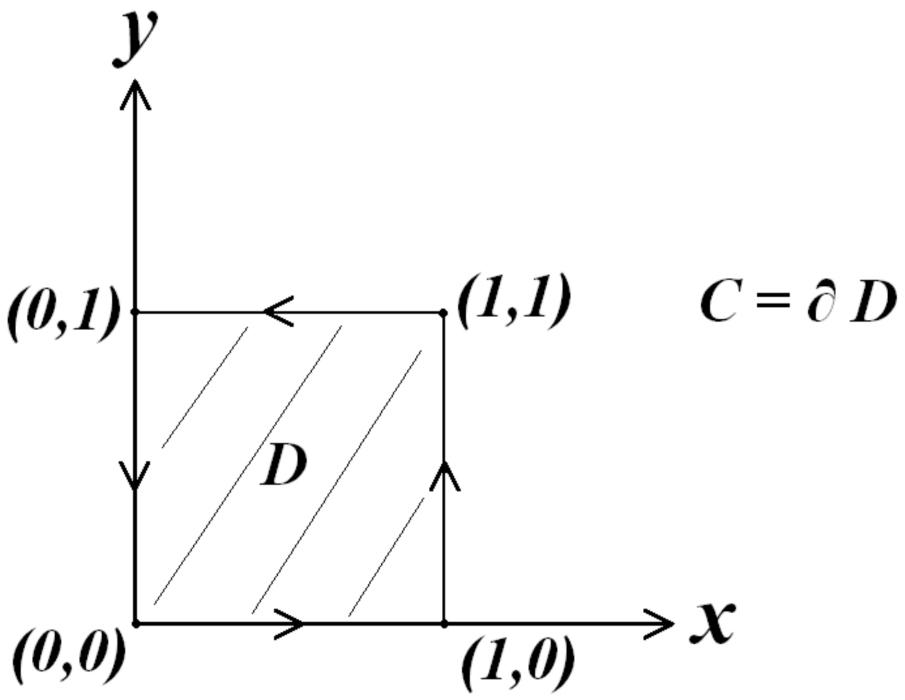


Example 24.3

Redo Example (17.2) in line integrals using Green's theorem. Evaluate

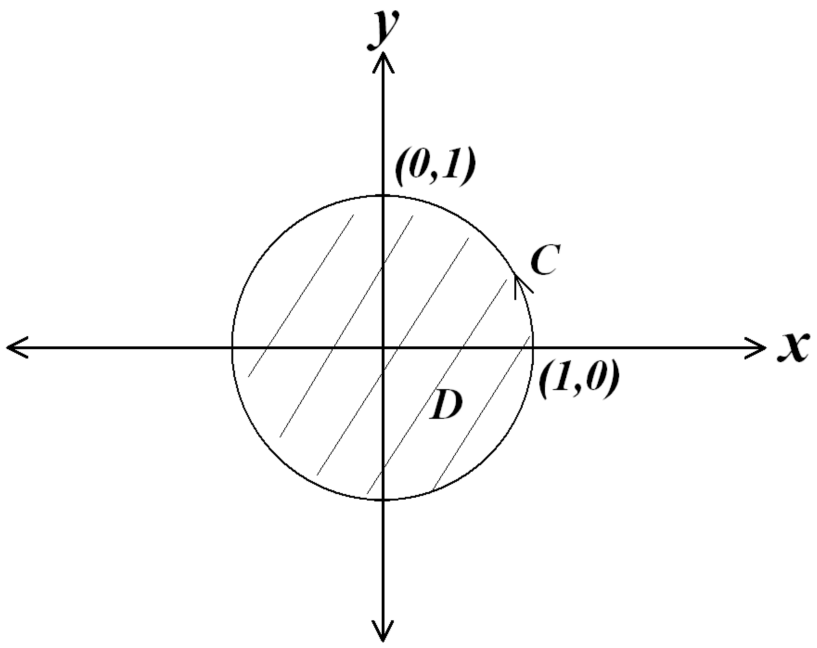
$$\int_C x^2\,dx + xy\,dy$$

over the unit square.



Example 24.4

Let $\mathbf{F}(x, y) = \left(2y + e^x,\ x + \sin(y^2)\right)$. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{s}$ where C is the unit circle centred at $(0, 0)$, traversed anticlockwise.



Note

The line integral is too hard to compute directly. Let's try using Green's theorem.

Example 24.5

Evaluate $\int_C \mathbf{F} \cdot d\mathbf{s}$ where

$$\mathbf{F}(x, y) = \left(\frac{-y}{x^2 + y^2}, \frac{x}{x^2 + y^2} \right)$$

and C is the curve $3x^2 + y^2 = 1$ oriented anticlockwise.

Note

Direct calculation of the line integral along the ellipse C is too hard.

Area of region in the xy-plane

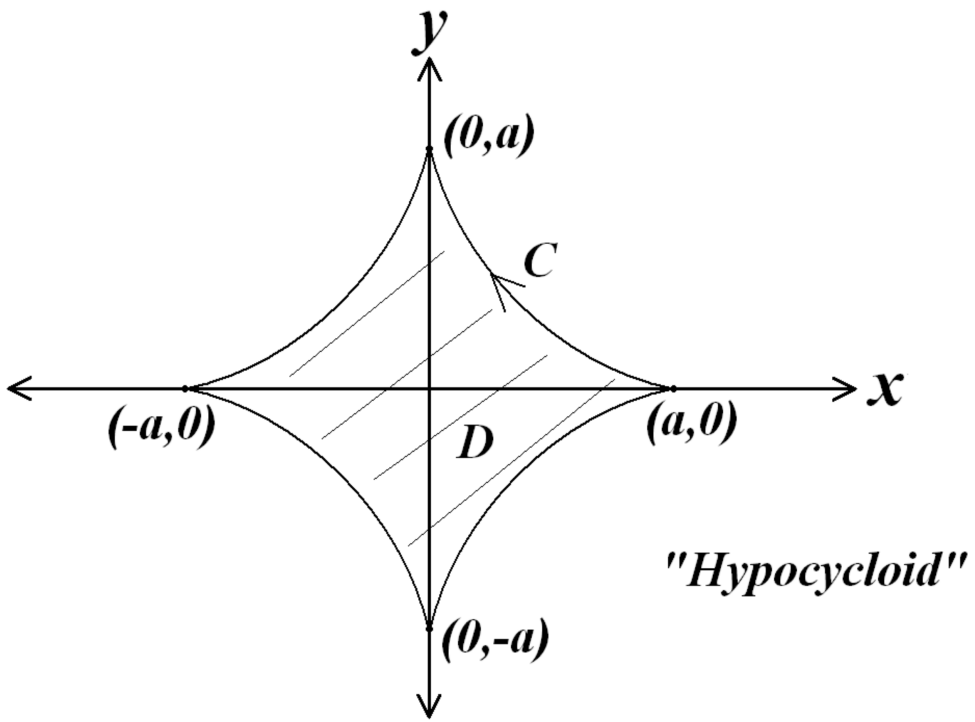
If C is a simple closed curve oriented in the positive direction that bounds a region D, then

$$\boxed{\text{Area of } D \;= \frac{1}{2}\int_{C=\partial D} x\,dy - y\,dx}$$

Example 24.6

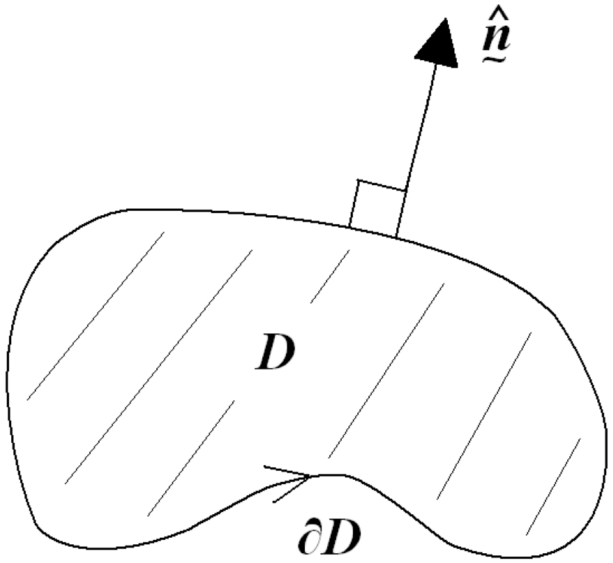
Find the area of the region enclosed by

$$x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}, \ (a > 0).$$



6.25. Divergence theorem in the plane

Let D be a simply connected region in the xy-plane bounded by a piecewise smooth simple closed curve $C = \partial D$ with positive orientation (anticlockwise). Let $\hat{\mathbf{n}}$ be the unit outward normal vector to ∂D in the xy-plane.



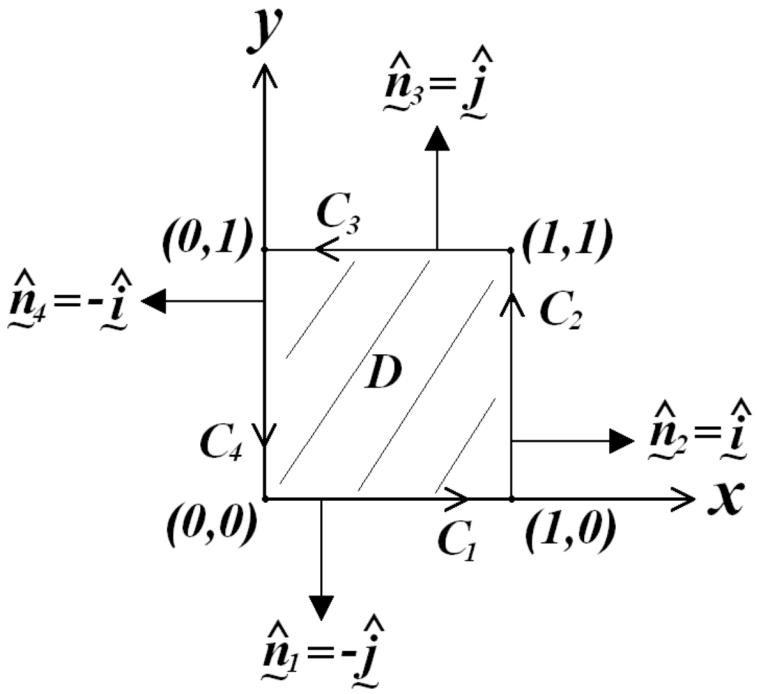
Let $\mathbf{F} = P(x, y)\,\mathbf{i} + Q(x, y)\,\mathbf{j}$ be a C^1 vector field on an open region containing D. Then

$$\int_{C=\partial D} \mathbf{F} \cdot \hat{\mathbf{n}} = \iint_D \nabla \cdot \mathbf{F}\, dx\, dy = \iint_D \operatorname{div} \mathbf{F}\, dx\, dy$$

This is another form of Green's theorem.

Example 25.1

Let $\mathbf{F}(x, y) = y^3\,\mathbf{i} + x^5\,\mathbf{j}$ and consider the region D bounded by the square with vertices $(0,0), (1,0), (1,1), (0,1)$. Evaluate $\int_{\partial D} \mathbf{F} \cdot \hat{\mathbf{n}}\, ds$ directly and also via the Divergence theorem in the plane.

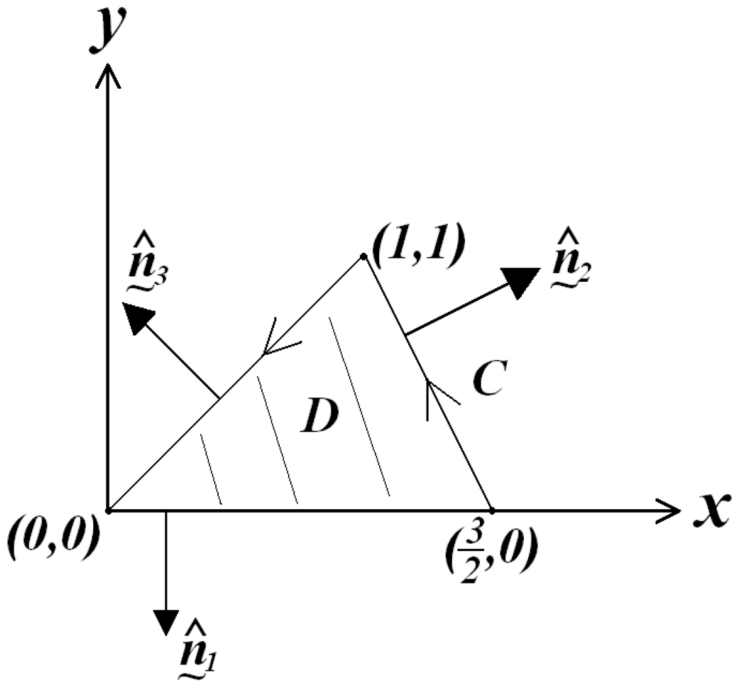


Example 25.2

Let C be the triangle with vertices $(0,0)$, $\left(\frac{3}{2},0\right)$ and $(1,1)$ traversed anticlockwise.

Evaluate $\displaystyle\int_C \mathbf{F}\cdot\hat{\mathbf{n}}\,ds$ where

$$\mathbf{F}(x,y) = (2x^2y - 3x + \sin(5y), 5y - 2xy^2 - \cos^3(4x)).$$



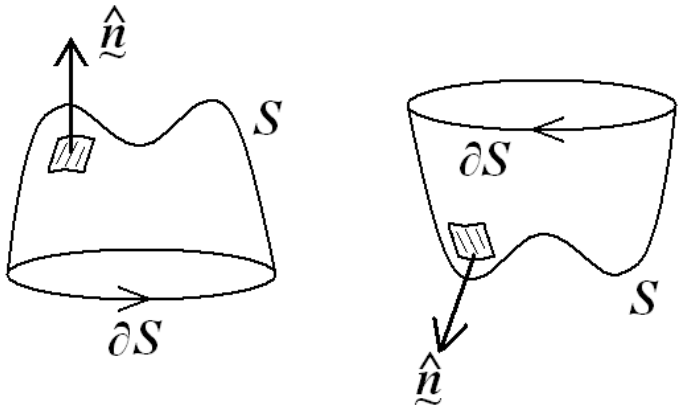
Note

Calculation of the path integral around C is quite complicated. Let's use the Divergence theorem in the plane.

6.26. Stokes' Theorem

Let S be an oriented surface and $C = \partial S$ its boundary curve.

A given orientation $\hat{\mathbf{n}}$ of S induces a positive orientation of C: if you walk in the positive direction around C with your head pointing in the direction of $\hat{\mathbf{n}}$, then S will always be on your left.



Note that the orientation of C and the unit normal $\hat{\mathbf{n}}$ of S are related by the right-hand rule.

Think of S as a fishing net and $C = \partial S$ as its rim.

Theorem 26.1 (Stokes' theorem)

Let S be an oriented (piecewise smooth) surface, with orientation given by $\hat{\mathbf{n}}$, bounded by a simple closed (piecewise smooth) curve $C = \partial S$ with induced positive orientation. Let $\mathbf{F}$ be a C^1 vector field on an open region of $\mathbb{R}^3$ containing S. Then

$$\iint_S (\boldsymbol{\nabla} \times \mathbf{F}) \cdot d\mathbf{S} = \iint_S (\operatorname{curl} \mathbf{F} \cdot \hat{\mathbf{n}})\, dS = \int_{C=\partial S} \mathbf{F} \cdot d\mathbf{c}.$$

Note

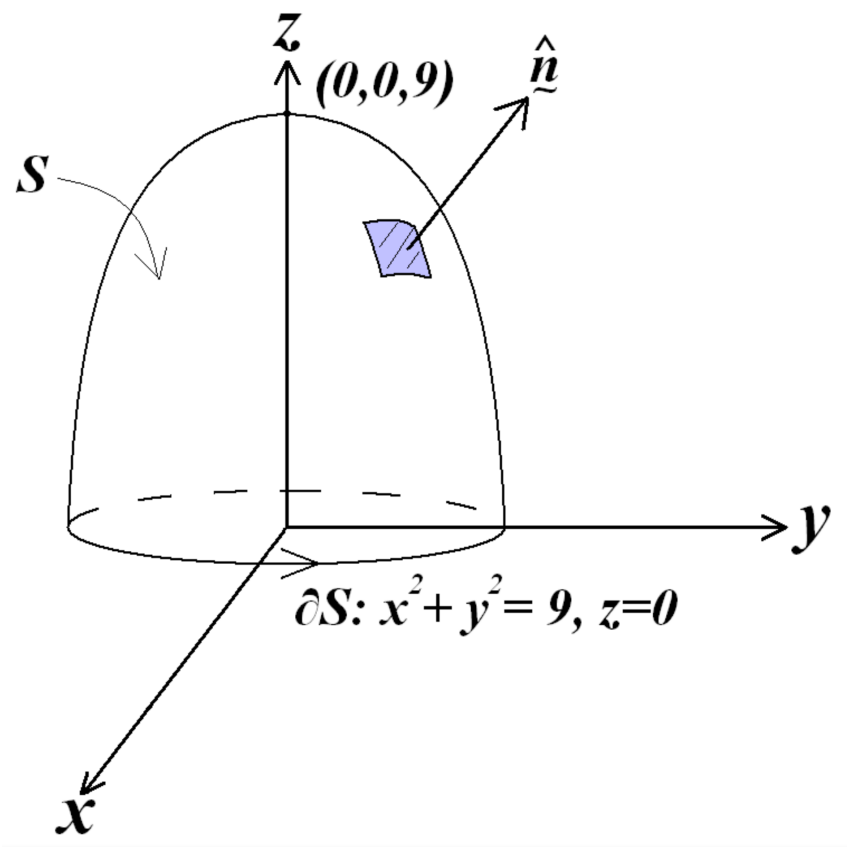
Green's theorem is a special case of Stokes' theorem, where S and ∂S are confined to the xy-plane.

Example 26.2

Let

$$\mathbf{F}(x, y, z) = (2z - y)\,\mathbf{i} + (x + z)\,\mathbf{j} + (3x - 2y)\,\mathbf{k}\,.$$

and consider the the paraboloid S given by $z = 9 - x^2 - y^2,\ z \geq 0$. Evaluate $\iint\limits_S (\nabla \times \mathbf{F}) \cdot\ d\mathbf{S}$ directly and also via Stokes' theorem.



Example 26.3

Evaluate

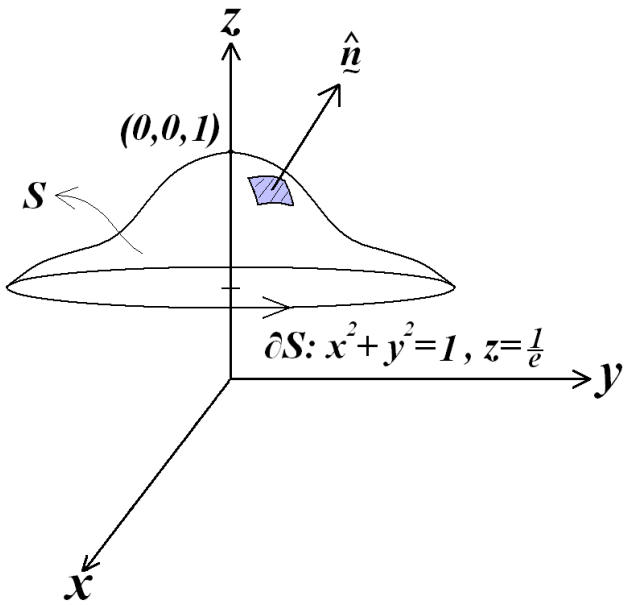
$$\iint_S (\nabla \times \mathbf{F}) \cdot \, d\mathbf{S}$$

where

$$\mathbf{F}(x, y, z) = \left(e^{y+z} - 2y, \; xe^{y+z}, \; e^{x+y}\right)$$

and S is the surface of the bell

$$z = e^{-(x^2+y^2)} \quad \text{for } z \geq \frac{1}{e}.$$

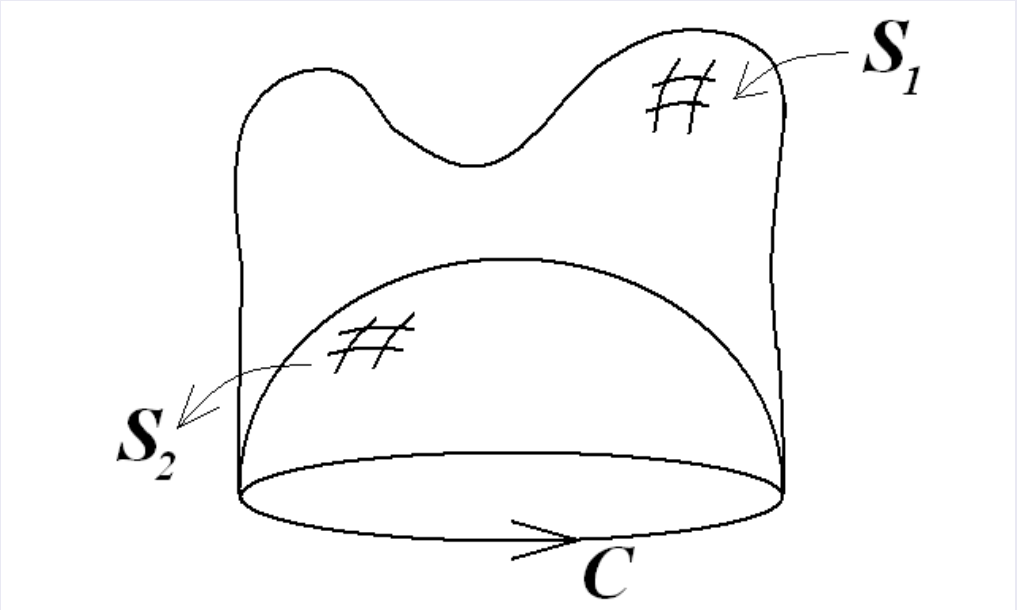


Note

For any two surfaces S_1, S_2 with the same boundary C, Stokes' theorem implies that

$$\iint_{S_1} (\nabla \times \mathbf{F}) \cdot \, d\mathbf{S} = \iint_{S_2} (\nabla \times \mathbf{F}) \cdot \, d\mathbf{S}$$

$$= \int_C \mathbf{F} \cdot \, d\mathbf{s}$$



So try applying Stokes' theorem and use the simplest surface with consistent orientation with C.

Note

If the surface is closed (i.e., there is no boundary), Stokes' theorem implies

$$\iint_S (\nabla \times \mathbf{F}) \cdot \, d\mathbf{S} = 0.$$

Example 26.4

Show that $\iint_S (\nabla \times \mathbf{F}) \cdot \, d\mathbf{S} = 0$ if S is a sphere of radius a, centred at $(0, 0, 0)$.

6.27. Conservative Fields

Let $\mathbf{F}$ be a C^1 vector field defined on $\mathbb{R}^2$ or $\mathbb{R}^3$.
The following conditions are all equivalent.

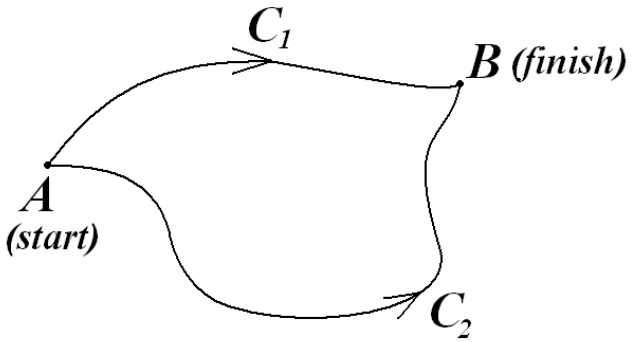
1. For any oriented simple closed curve C,

$$\int_C \mathbf{F} \cdot \, d\mathbf{c} = 0.$$

2. For any two oriented simple curves C_1 and C_2 with the same endpoints,

$$\int_{C_1} \mathbf{F} \cdot \, d\mathbf{c} = \int_{C_2} \mathbf{F} \cdot \, d\mathbf{c}.$$

(path independence)



3. $\mathbf{F} = \nabla\phi$ for some scalar function ϕ. ($\mathbf{F}$ is a gradient field).
4. $\nabla \times \mathbf{F} = \mathbf{0}$. ($\mathbf{F}$ is an irrotational field).

A vector field satisfying one (and hence all) of the four conditions is called a conservative vector field.

We will prove that (3) implies (2) in the above theorem, i.e., if $\mathbf{F} = \boldsymbol{\nabla}\phi$, then $\int_C \mathbf{F} \cdot d\mathbf{c}$ depends only on the endpoints of C.

Example 27.1

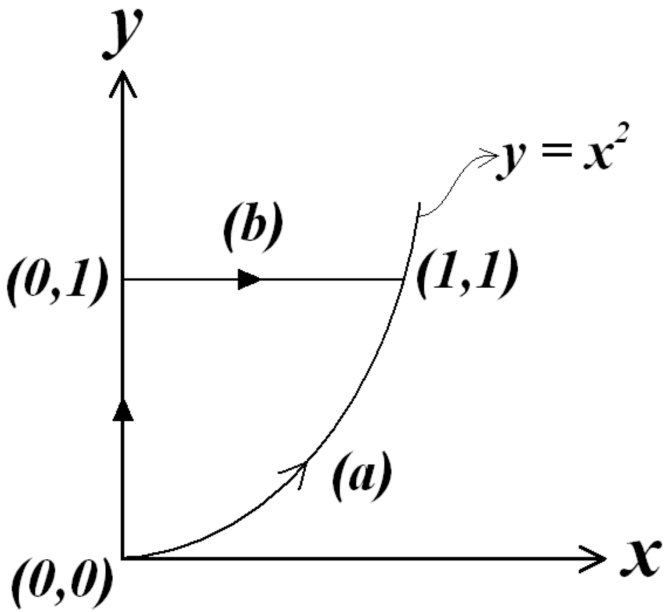
Let $\mathbf{F}(x, y) = x\,\mathbf{i} + y\,\mathbf{j}$.

1. Show that F is conservative.
2. Determine the scalar potential ϕ for $\mathbf{F}$.

Example 27.2

Continue from the previous example.

3. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{c}$ along C given by
   (a) $y = x^2$ from $(0,0)$ to $(1,1)$;
   (b) the line segments joining $(0,0)$ to $(0,1)$ to $(1,1)$.
4. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{c}$ where C is the unit circle centred at $(0,0)$.



Example 27.3

Determine the work done by

$$\mathbf{F}(x, y, z) = (x^2,\ \cos y \sin z,\ \sin y \cos z)$$

to move a particle along the path

$$\mathbf{c}(t) = (t^2 + 1,\ e^t,\ e^{2t}),\ 0 \leq t \leq 1.$$

Note

Calculation of the line integral is too hard, but we notice that $\mathbf{F}$ is a conservative field.

6.28. Gauss' Divergence Theorem

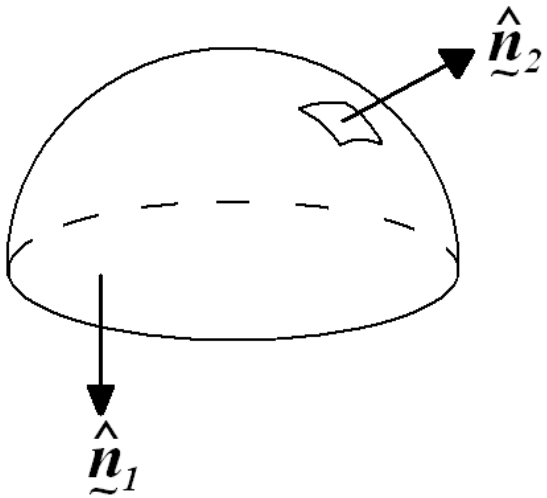
Theorem 28.1 (The divergence theorem)

Let Ω be a simple solid region in $\mathbb{R}^3$ bounded by a piecewise smooth orientable closed surface $S = \partial\Omega$. If $\mathbf{F}$ is a C^1 vector field in an open set containing Ω, then

$$\iiint_{\Omega} \nabla \cdot \mathbf{F}\, dV = \iiint_{\Omega} \operatorname{div} \mathbf{F}\, dV = \iint_{\partial\Omega} \mathbf{F} \cdot\, d\mathbf{S} = \iint_{\partial\Omega} \mathbf{F} \cdot \hat{\mathbf{n}}\, dS$$

where $\hat{\mathbf{n}}$ is the outward unit normal field for the surface $\partial\Omega$.

Example: solid hemisphere (bounded by 2 smooth surfaces)



Example 28.2

Redo Example (23.1) of surface integrals using Gauss' Divergence theorem. Evaluate

$$\iint_S \mathbf{F} \cdot d\mathbf{S}$$

where $\mathbf{F}(x, y, z) = x\,\mathbf{i} + y\,\mathbf{j} + z\,\mathbf{k}$ and S is the unit sphere centred at origin, oriented with outward unit normals.

Example 28.3

Redo Example (23.2) of surface integrals using Gauss' Divergence theorem. Evaluate

$$\iint_S (x^3\,\mathbf{i} + y^3\,\mathbf{j}) \cdot \, d\mathbf{S}$$

where S is the closed cylinder $x^2 + y^2 = 4, z = 0, \; z = 5$, oriented with outward unit normals.

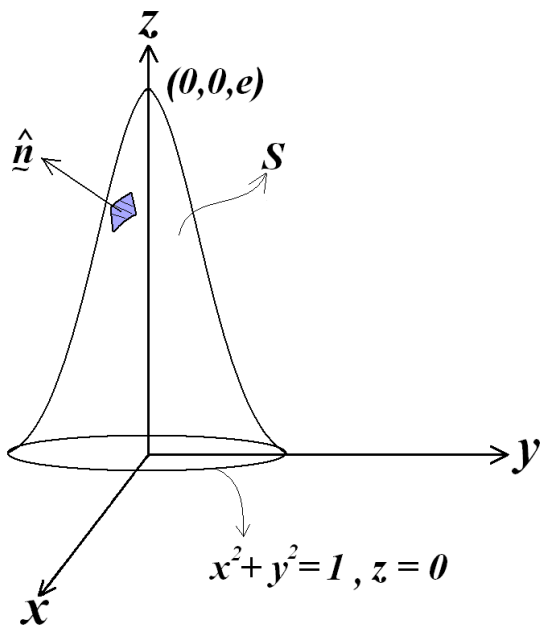
Example 28.4

Evaluate $\iint_S \mathbf{F} \cdot d\mathbf{S}$ where S is the surface of the bell

$$z = (1 - x^2 - y^2)e^{(1-x^2-3y^2)}$$

for $z \geq 0$, oriented with outward unit normals, and

$$\mathbf{F}(x, y, z) = \left(e^y \cos z,\ \sqrt{x^3 + 1} \sin z,\ x^2 + y^2 + 3\right).$$



Note

Evaluating the surface integral over the bell directly is hard.

Volume via Surface Integrals

If $\mathbf{F}(x, y, z) = x\,\mathbf{i} + y\,\mathbf{j} + z\,\mathbf{k}$ and Ω is a region to which Gauss' theorem applies, then

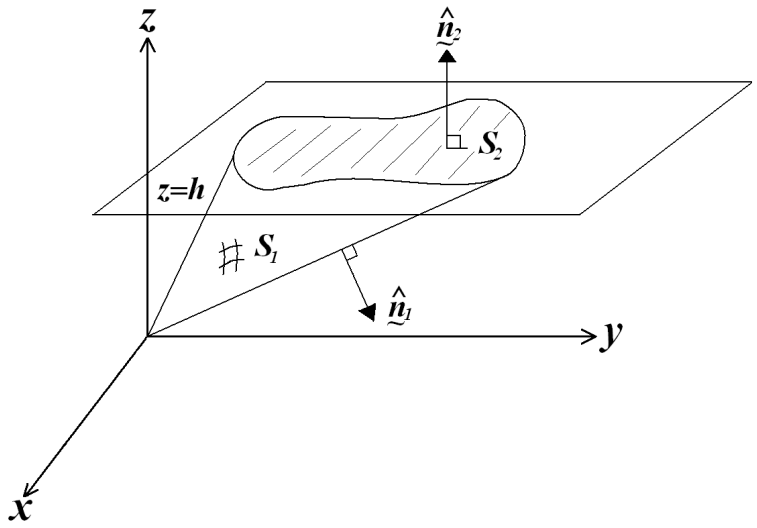
$$\boxed{\text{volume of } \Omega = \frac{1}{3} \iint_{\partial\Omega} \mathbf{F} \cdot \, d\mathbf{S}}$$

Proof.

$\square$

Example 28.5

Find the volume of a solid cone with base area A and height h.



6.29. Applications of integral theorems

Example 29.1 (Gauss' Law)

Let Ω be an elementary region in $\mathbb{R}^3$ bounded by the closed surface $\partial\Omega$. Then if $\mathbf{0} \notin \partial\Omega$,

$$\iint_{\partial\Omega} \frac{\mathbf{r}}{r^3} \cdot \, d\mathbf{S} = \begin{cases} 0, & \mathbf{0} \notin \Omega \\ 4\pi, & \mathbf{0} \in \Omega \end{cases}$$

where $\mathbf{r} = x\,\mathbf{i} + y\,\mathbf{j} + z\,\mathbf{k}$ and $r = \sqrt{x^2 + y^2 + z^2}$.

Example 29.2 (Continuity Equation for Fluid Flow)

A fluid of density $\rho(\mathbf{r}, t)$ moves with velocity $\mathbf{v}(\mathbf{r}, t)$. Let $\boldsymbol{J} = \rho\mathbf{v}$ (the fluid current). By the *law of conservation of mass* for ρ and $\mathbf{v}$ we mean that the conservation equation

$$\frac{\partial}{\partial t} \iiint_{\Omega} \rho(\mathbf{r}, t)\, dV = -\iint_{\partial\Omega} \boldsymbol{J} \cdot \mathbf{n}\, dS \tag{46}$$

holds for all solid region Ω in $\mathbb{R}^3$ (where $\mathbf{n}$ is the outward normal). Show that the conservation equation (46) is equivalent to

$$\frac{\partial \rho}{\partial t} = -\boldsymbol{\nabla} \cdot \boldsymbol{J}\,.$$

This is called the equation of continuity.

Example 29.3

Maxwell's Equations for Electromagnetic Fields.
Define the following quantities

- electric charge density $\rho(\mathbf{r}, t)$
- electric current $\boldsymbol{J}(\mathbf{r}, t)$
- vector field for magnetic force $\boldsymbol{B}(\mathbf{r}, t)$
- vector field for electric force $\boldsymbol{E}(\mathbf{r}, t)$
- permittivity of free space ε_0
- permeability of free space μ_0.

In S.I. units, Maxwell's equations can be written as

(a) $\boldsymbol{\nabla} \times \boldsymbol{E} = -\dfrac{\partial \boldsymbol{B}}{\partial t}$ (Faraday's Law).
(If $\boldsymbol{B}$ changes with time, then an electric field is produced)

(b) $\boldsymbol{\nabla} \cdot \boldsymbol{E} = \dfrac{\rho}{\varepsilon_0}$ (Gauss' Law).
(Charges present make $\boldsymbol{\nabla} \cdot \boldsymbol{E}$ nonzero.)

Example 29.4

(c) $\boldsymbol{\nabla} \times \boldsymbol{B} = \mu_0 \boldsymbol{J} + \varepsilon_0 \mu_0 \dfrac{\partial \boldsymbol{E}}{\partial t}$ (Ampere's Law).

(If $\boldsymbol{\nabla} \times \boldsymbol{B} \neq \boldsymbol{0}$ then there are currents or $\boldsymbol{E}$ changes with time.)

(d) $\boldsymbol{\nabla} \cdot \boldsymbol{B} = 0$.

($\boldsymbol{B}$ is always incompressible. There are no magnetic sources.)

Derive some consequences of Maxwell's equations.

Note

This is the wave equation in a vacuum. The wave equation can be written in the form

$$\nabla^2 \boldsymbol{E} = \frac{1}{c^2}\frac{\partial^2 \boldsymbol{E}}{\partial t^2}$$

where c is the speed of the wave.
Note that the permittivity ε_0 of free space is

$$8.85418782 \times 10^{-12}\ \mathrm{m}^{-3}\ \mathrm{kg}^{-1}\ \mathrm{s}^4\ \mathrm{A}^2$$

and the permeability μ_0 of free space is

$$1.25663706 \times 10^{-6}\ \mathrm{m}\ \mathrm{kg}\ \mathrm{s}^{-2}\ \mathrm{A}^{-2}.$$

The electromagnetic waves travel with speed

$$\frac{1}{\sqrt{\varepsilon_0 \mu_0}} \approx 300,000\ \mathrm{km/sec}.$$

This was predicted by Maxwell 20 years before experimentally observed by Hertz.